

POSITIVE GENERALIZED CORNER-VECTOR DESIGNS ON $\mathbb{S}^3$ HAVE DEGREE AT MOST 13

ABSTRACT. Tanino, Tamaru, Hirao, and Sawa have proved that positive weighted spherical designs obtained from generalized corner-vector orbits in dimension four have degree at most 15, and asked whether a weighted 15-design on $\mathbb{S}^3$ with more than five proper orbits exists. We give a B_4 -invariant polynomial of degree 14 that is positive on every generalized corner-vector orbit but has negative spherical mean. Consequently, every positive design of this type on $\mathbb{S}^3$ has degree at most 13. This gives a negative answer, for positive weights, to their Problem 6.1, independently of the number of proper orbits.

1. THE OBSTRUCTION

Let B_4 act on $\mathbb{R}^4$ by signed coordinate permutations. For $a > 0$ and $s \in \{0, 1, 2, 3\}$, set

$$v_{a,s} = \frac{1}{\sqrt{a^2 + s}}(a, \underbrace{1, \dots, 1}_s, 0, \dots, 0) \in \mathbb{S}^3,$$

and denote its B_4 -orbit by $v_{a,s}^{B_4}$. These are the generalized corner-vector orbits of Tanino–Tamaru–Hirao–Sawa [1, Definition 2.19]; the orbit is called proper when $a \neq 1$. They proved that a positive weighted design assembled from such orbits has degree at most 15 in dimension four [1, Theorem 3.1], and they asked [1, Problem 6.1]:

Does there exist a weighted 15-design on $\mathbb{S}^3$ of type (1) with more than 5 proper orbits?

Their “type (1)” is the display numbered (1) on p. 1388 of [1]: a cubature formula for normalized surface measure whose node set is a union of generalized corner-vector orbits, carrying one weight per orbit. That is the class of formula (1) below, and it is also the class occurring in the hypothesis of [1, Theorem 3.1], whose formula (11) has the same shape. Let σ denote normalized surface measure on $\mathbb{S}^3$.

Theorem 1. *Let $m \geq 1$, $a_i > 0$, $s_i \in \{0, 1, 2, 3\}$, and $W_i > 0$. The formula*

$$(1) \quad \int_{\mathbb{S}^3} f d\sigma = \sum_{i=1}^m W_i \sum_{x \in v_{a_i, s_i}^{B_4}} f(x)$$

cannot be exact for every polynomial f of degree at most 14.

2020 *Mathematics Subject Classification.* Primary 65D32; Secondary 05B30.

Key words and phrases. spherical design, cubature formula, hyperoctahedral group, generalized corner vector, separating polynomial, Bernstein basis.

Corollary 2. *Every positive weighted spherical design of generalized corner-vector type on $\mathbb{S}^3$ has degree at most 13. In particular, Problem 6.1 of [1] has a negative answer for positive weights, as required of a weighted design by [1, Definition 2.2], regardless of the number of proper orbits.*

Put $y_i = x_i^2$, and let $e_j = e_j(y_1, y_2, y_3, y_4)$ be the j th elementary symmetric polynomial. Define

$$(2) \quad p = 2 + 240e_2 + 1345e_3 + 227924e_4 - 3518e_2^2 - 448e_2e_3 + 283604e_3^2 \\ - 1529173e_2e_4 + 12886e_2^3 + 4855151e_3e_4 - 132972e_2^2e_3.$$

Then p is B_4 -invariant and has degree 14 in $x_1, \dots, x_4$. For a polynomial f on $\mathbb{S}^3$, write

$$\langle f \rangle = \int_{\mathbb{S}^3} f d\sigma.$$

Lemma 3. *The spherical mean of p is negative:*

$$\langle p \rangle = -\frac{60763}{143360} < 0.$$

Proof. For a uniformly distributed point of $\mathbb{S}^3$, the vector (y_1, y_2, y_3, y_4) has the Dirichlet distribution with parameters $(1/2, 1/2, 1/2, 1/2)$. Hence, for $\alpha_i \in \mathbb{Z}_{\geq 0}$,

$$\langle y_1^{\alpha_1} \dots y_4^{\alpha_4} \rangle = \frac{\prod_{i=1}^4 (1/2)^{\alpha_i}}{(2)_{\alpha_1 + \dots + \alpha_4}}, \quad (z)_0 = 1.$$

Expanding (2) and applying this identity gives the stated value. $\square$

Lemma 4. *For every $a > 0$ and $s \in \{0, 1, 2, 3\}$,*

$$p(v_{a,s}) > 0.$$

Proof. For $s = 0$, one has $v_{a,0} = (1, 0, 0, 0)$ and $p(v_{a,0}) = 2$. Let $s \in \{1, 2, 3\}$ and set

$$u = \frac{a^2}{a^2 + s} \in (0, 1), \quad c = \frac{1 - u}{s}.$$

Since p is a symmetric function of the y_i , we may take the squared coordinates of $v_{a,s}$ to be $(u, c, \dots, c, 0, \dots, 0)$, with c repeated s times. Direct substitution into (2) gives

$$p(v_{a,1}) = H_1(u), \quad 32p(v_{a,2}) = H_2(u), \quad 729p(v_{a,3}) = H_3(u),$$

where

$$H_1(u) = -12886u^6 + 38658u^5 - 42176u^4 + 19922u^3 - 3758u^2 + 240u + 2, \\ H_2(u) = -598374u^7 + 2387827u^6 - 3979288u^5 + 3540713u^4 \\ - 1750294u^3 + 440357u^2 - 42268u + 1391, \\ H_3(u) = -51606520u^7 + 213893327u^6 - 355558770u^5 + 299081978u^4 \\ - 130946506u^3 + 26917938u^2 - 1819702u + 39713.$$

We certify positivity by exact Bernstein coefficients. For an interval $I = [r, q] \subseteq [0, 1]$ and a polynomial H of degree d , write

$$H(r + (q - r)z) = \sum_{k=0}^d \beta_{I,k}(H) \binom{d}{k} z^k (1 - z)^{d-k}.$$

If $H(r + (q - r)z) = \sum_{j=0}^d c_j z^j$, then

$$(3) \quad \beta_{I,k}(H) = \sum_{j=0}^k c_j \frac{\binom{k}{j}}{\binom{d}{j}}.$$

For each s , the following table gives a subdivision $0 = r_0 < r_1 < r_2 < r_3 = 1$ and the minimum Bernstein coefficient on each successive interval, where $d = \deg H_s$, so that $d = 6$ for $s = 1$ and $d = 7$ for $s = 2, 3$. The coefficients (3) depend on d ; in particular H_1 is not to be padded to degree 7.

s	(r_0, r_1, r_2, r_3)	$(\min_k \beta_{[r_{j-1}, r_j], k}(H_s))_{j=1}^3$
1	$(0, \frac{1}{6}, \frac{5}{6}, 1)$	$(\frac{14287}{7776}, \frac{46531}{23328}, \frac{14287}{7776})$
2	$(0, \frac{1}{12}, \frac{3}{4}, 1)$	$(\frac{137627845}{2985984}, \frac{414113333}{5971968}, \frac{219941}{5376})$
3	$(0, \frac{1}{20}, \frac{3}{5}, 1)$	$(\frac{542097724127}{448000000}, \frac{17317137}{15625}, \frac{4479262}{3125})$

Every displayed number is positive. Since the Bernstein basis functions are nonnegative on $[0, 1]$ and sum to 1, it follows that $H_s(u) > 0$ on every listed interval, hence on $[0, 1]$. Therefore $p(v_{a,s}) > 0$ for $s = 1, 2, 3$ as well. $\square$

Proof of Theorem 1. Assume that (1) is exact through degree 14. Applying it to the polynomial p and using its B_4 -invariance gives

$$-\frac{60763}{143360} = \langle p \rangle = \sum_{i=1}^m W_i |v_{a_i, s_i}^{B_4}| p(v_{a_i, s_i}).$$

The right-hand side is strictly positive by $W_i > 0$ and Lemma 4, contradicting Lemma 3. $\square$

Remark 5. Since $-I \in B_4$, every formula of the form (1) is antipodal. Exactness through degree 14 is therefore equivalent to exactness through degree 15.

Positivity is essential, and the positive-weight reading is the one intended in [1]: a weighted design there carries by definition a weight function with values in $\mathbb{R}_{>0}$ [1, Definition 2.2], so the hypothesis $W_i > 0$ of Theorem 1 is theirs and not a restriction added here. The displays [1, (1) and (11)] do not repeat the sign condition; that the positive reading is nonetheless the intended one is confirmed by [1, §6], where it is recorded, immediately before Problem 6.1 is posed, that 15-designs on $\mathbb{S}^3$ with negative weights do exist. Theorem 1 accordingly says nothing about signed cubature formulas,

nor about node families outside the generalized corner-vector construction. Degree 13 is attained within this class: [1, §6] reports two weighted 13-designs on $\mathbb{S}^3$ of this type, each with five orbits and positive weights, with nodes and weights given there numerically to six significant figures. On the evidence of those examples the bound of Corollary 2 is sharp; verifying them exactly is not attempted here.

Exact verification. Everything above is finite and exact, and is recomputable from the data printed here; the only mathematical input is the eleven coefficients of (2). From them one expands p into its 130 monomials in the y_i and confirms its B_4 -invariance and its degree 14 in $x_1, \dots, x_4$; evaluates $\langle p \rangle = -60763/143360$ by two routes, direct expansion against the Dirichlet moment functional of Lemma 3 and a harmonic route through the Laplacian, the moment functional itself being cross-validated against the 24-cell through degree 5 and against an iterated-Laplacian formula on all 330 even monomials of degree at most 14; substitutes the profiles of Lemma 4 to recover H_1, H_2, H_3 with their integrality scalings 1, 32, 729; verifies the identity (3) itself, rather than assuming it, and with it all nine minimum Bernstein coefficients of the table; and re-certifies positivity of H_1, H_2, H_3 on $[0, 1]$ a second way, by exact Sturm root counting. No floating-point arithmetic enters at any stage. This computation was re-run independently in exact rational arithmetic, and all 48 of its checks agree with the numbers printed here. The program and its output log accompany this note; nothing above depends on them, since the Bernstein table is itself the certificate.

Scope. Two limits belong with that statement. First, the claim quantifies over the continuum $a > 0$, so no census is possible and none was attempted: the universal statement is discharged exactly by the Bernstein certificate over the tabulated partition, re-certified by Sturm, with exact evaluation of p at 609 genuine orbit points as a further check. Second, agreement is what is claimed for the independent re-run, not a second proof: a few of its checks are self-tests of its own construction, which no object could fail, and the agreement of the Pochhammer and double-factorial forms of the Dirichlet moment is an algebraic rewriting rather than an independent derivation.

REFERENCES

- [1] K. Tanino, T. Tamaru, M. Hirao, and M. Sawa, More on the corner-vector construction for spherical designs, *Algebraic Combinatorics* **8** (2025), no. 5, 1387–1414, doi:10.5802/alco.441.